

Maven Project To Connect To Web Servers Using Selenium Dependency

Step1:

Create a maven project on the terminal with the command:

```
$mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp04 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

```
vaishnavi@vaishnavi-virtualbox:~/devops215$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenChromeApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> mvn:archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< mvn:archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- mvn:archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/vaishnavi/devops21
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenChromeApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/vaishnavi/devops21/MyMavenChromeApp
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.065 s
[INFO] Finished at: 2020-03-25T13:11:36+05:30
[INFO]
vaishnavi@vaishnavi-virtualbox:~/devops215$
```

The command can be broken down as:

- DgroupId=com.example – project’s base package (reverse-domain style).
- DartifactId=MyMavenSeleniumApp04 – defines the name of the project (will also be the folder name).
- DarchetypeArtifactId=maven-archetype-quickstart – sets the Maven template to be the basic Java application.
- DinteractiveMode=false – deactivates interactive mode and uses the given values automatically.

Step 2:

Use \$cd command to go into the directory.

```
$gedit pom.xml
```

pom.xml file is created to configure the Maven project. It defines the project information, adds required dependencies like Guava and Commons IO libraries, and includes build plugins to compile the Java code, run unit tests, and package the application into an executable JAR file.

```
<dependencies>
```

```
<dependency>
```

```
<groupId>junit</groupId>
```

```
<artifactId>junit</artifactId>
```

```
<version>3.8.1</version>
```

```
<scope>test</scope>

</dependency>

<!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
<dependency>
  <groupId>org.seleniumhq.selenium</groupId>
  <artifactId>selenium-java</artifactId>
  <version>4.29.0</version>
</dependency>
</dependencies>

<!-- Build: Configuring plugins and build settings -->
<build>
  <plugins>
    <!-- Example: Maven Compiler Plugin to compile Java code -->
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.8.1</version>
      <configuration>
        <source>1.7</source>
        <target>1.7</target>
      </configuration>
    </plugin>
    <!-- Example: Maven Surefire Plugin to run tests -->
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-surefire-plugin</artifactId>
      <version>2.22.2</version>
    </plugin>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>

      <artifactId>maven-jar-plugin</artifactId>
```

```
<version>3.0.1</version>

<configuration>

  <archive>

    <manifestEntries>

      <Main-Class>com.example.App</Main-Class>

    </manifestEntries>

  </archive>

</configuration>

</plugin>
```

```
</plugins>
```

```
</build>
```

```
</project>
```

Step 3:

inspect the AutomationExercise webpage to find the element IDs required for automation.

Right-click on the webpage and select Inspect. The HTML code will be displayed. Click on the cursor icon (arrow symbol) available on the top-left of the inspect panel, then hover over the username field to identify its ID, which is user-name.

Similarly, check the search button (ID:submit_search). Make a note of these IDs, as they will be used in the App.java file.

Next,

Edit the App.java file

App.java

```
package com.example;
```

```
import org.openqa.selenium.By;
```

```
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.chrome.ChromeDriver;
```

```
/**
```

```
 * Hello world!
```

```
 *
```

```
 */
```

```
public class App
{
    public static void main( String[] args )
    {
        WebDriver driver=new ChromeDriver();

        driver.get("https://www.saucedemo.com/");
        driver.manage().window().maximize();
        driver.findElement(By.id("user-name")).sendKeys("standard_user");
        driver.findElement(By.id("password")).sendKeys("secret_sauce");
        driver.findElement(By.id("login-button")).click();

        driver.get("https://practicetestautomation.com/practice-test-login/");
        driver.manage().window().maximize();
        driver.findElement(By.id("username")).sendKeys("student");
        driver.findElement(By.id("password")).sendKeys("Password123");
        driver.findElement(By.id("submit")).click();

        driver.get("https://automationexercise.com/products");
        driver.manage().window().maximize();
        driver.findElement(By.id("search_product")).sendKeys("blue top");
        driver.findElement(By.id("submit_search")).click();
    }
}
```

Execute command \$tree

```

vaishnavi@vaishnavi-virtualbox:~/devops21/Chromeex$ ls
pom.xml src target
vaishnavi@vaishnavi-virtualbox:~/devops21/Chromeex$ tree
.
├── pom.xml
├── src
│   ├── main
│   │   ├── java
│   │   │   ├── com
│   │   │   │   └── example
│   │   │       └── App.java
│   └── test
│       ├── java
│       │   ├── com
│       │   │   └── example
│       │       └── AppTest.java
└── target
    ├── classes
    │   └── com
    │       └── example
    │           └── App.class
    ├── generated-sources
    │   └── annotations
    ├── generated-test-sources
    │   └── test-annotations
    ├── maven-archiver
    ├── pom.properties
    ├── maven-status
    │   └── maven-compiler-plugin
    │       └── compile
    │           ├── default-compile
    │           │   ├── createdFiles.lst
    │           │   └── inputFiles.lst
    │           └── testCompile
    │               ├── default-testCompile
    │               │   ├── createdFiles.lst
    │               │   └── inputFiles.lst
    └── MyMavenSeleniumApp04-1.0-SNAPSHOT.jar
        ├── surefire-reports
        │   ├── com.example.AppTest.txt
        │   └── TEST-com.example.AppTest.xml
        └── test-classes
            ├── com
            │   └── example
            │       └── AppTest.class

```

28 directories, 13 files
vaishnavi@vaishnavi-virtualbox:~/devops21/Chromeex\$

Step 4:

Build and run the maven application

Execute the command \$mvn clean install

```

vaishnavi@vaishnavi-virtualbox:~/devops21/Chromeex$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] ----- com.example.MyMavenSeleniumApp04 1.0-SNAPSHOT -----
[INFO] Building MyMavenSeleniumApp04 1.0-SNAPSHOT
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenSeleniumApp04 ---
[INFO] Deleting /home/vaishnavi/devops21/Chromeex/target
[INFO]
[INFO] --- maven-resources-plugin:2.8:resources (default-resources) @ MyMavenSeleniumApp04 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vaishnavi/devops21/Chromeex/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenSeleniumApp04 ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/vaishnavi/devops21/Chromeex/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.8:testResources (default-testResources) @ MyMavenSeleniumApp04 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vaishnavi/devops21/Chromeex/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenSeleniumApp04 ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/vaishnavi/devops21/Chromeex/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenSeleniumApp04 ---
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.003 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.1.1:jar (default-jar) @ MyMavenSeleniumApp04 ---
[INFO] Building jar: /home/vaishnavi/devops21/Chromeex/target/MyMavenSeleniumApp04-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenSeleniumApp04 ---
[INFO] Installing /home/vaishnavi/devops21/Chromeex/target/MyMavenSeleniumApp04-1.0-SNAPSHOT.jar to /home/vaishnavi/.m2/repository/com/example/MyMavenSeleniumApp04/1.0-SNAPSHOT/MyMavenSeleniumApp04-1.0-SNAPSHOT.jar
[INFO] Installing /home/vaishnavi/devops21/Chromeex/pom.xml to /home/vaishnavi/.m2/repository/com/example/MyMavenSeleniumApp04/1.0-SNAPSHOT/MyMavenSeleniumApp04-1.0-SNAPSHOT.pom
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 2.032 s
[INFO] Finished at: 2020-03-25T13:12:41+05:30
[INFO]
vaishnavi@vaishnavi-virtualbox:~/devops21/Chromeex$

```

If the command shows output as “build success”, it means that the project has compiled successfully

Step 5:

Execute \$ java -jar target/MyMavenSeleniumApp03-1.0-SNAPSHOT

If we get error in executing the above jar file do run the below command

\$mvn exec:java -Dexec.mainClass="com.example.App"

```

[INFO] Scanning for projects...
[INFO]
[INFO] -----  

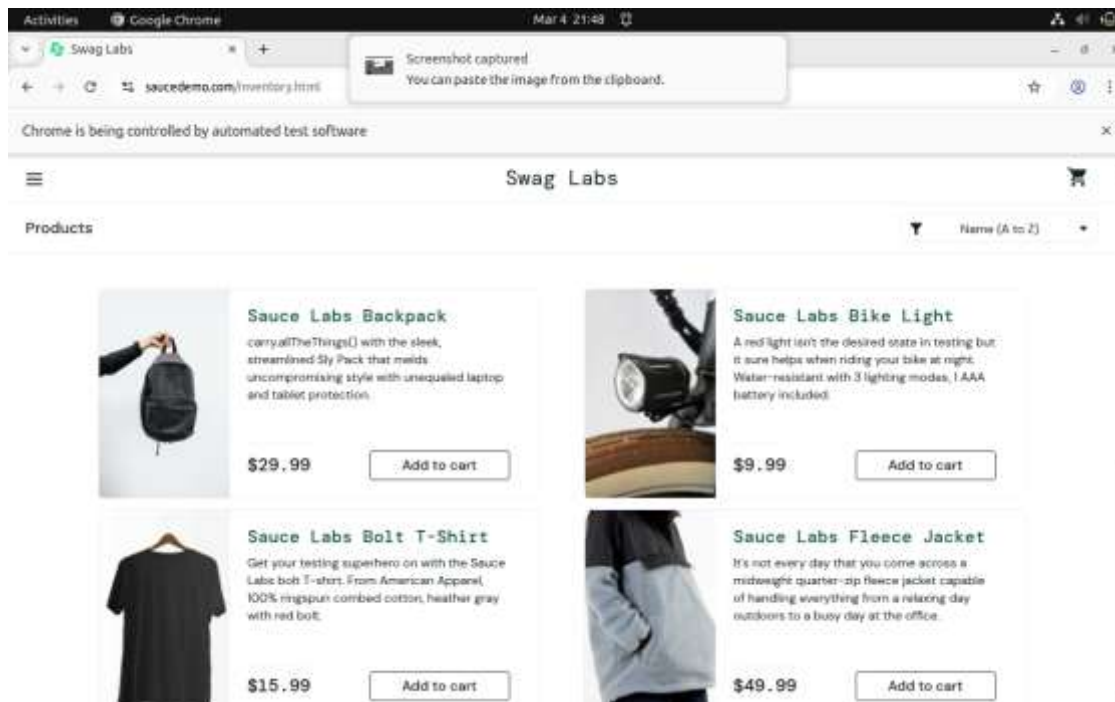
[INFO] Building MyMavenSeleniumApp03-1.0-SNAPSHOT  

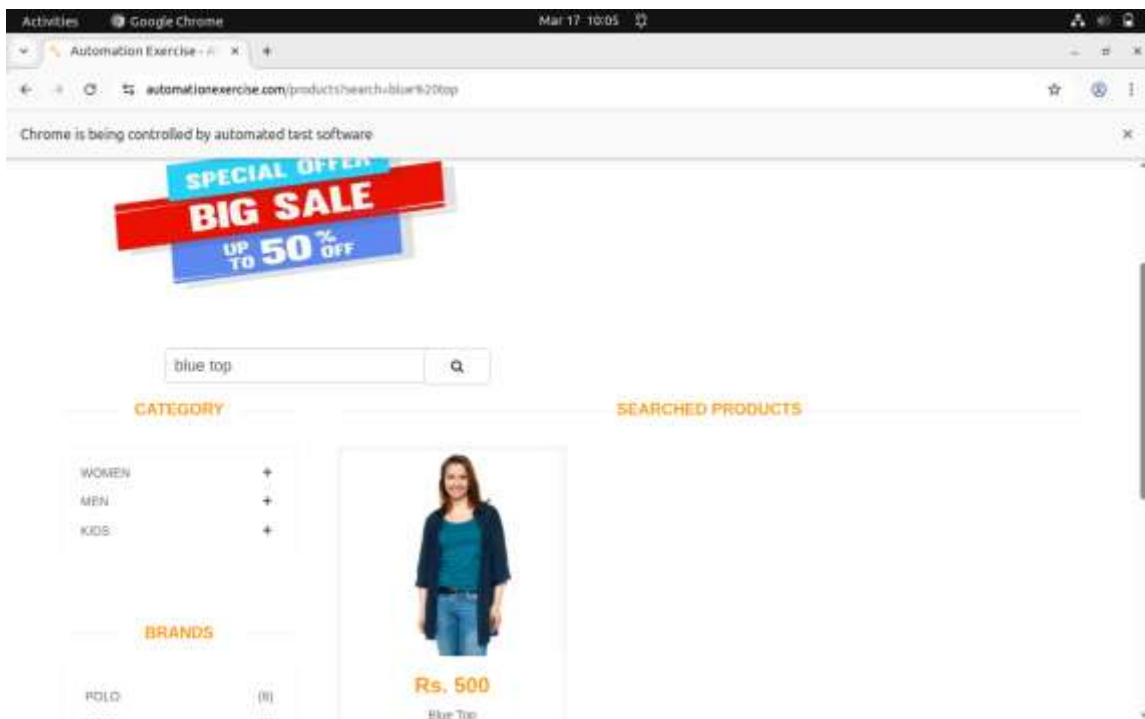
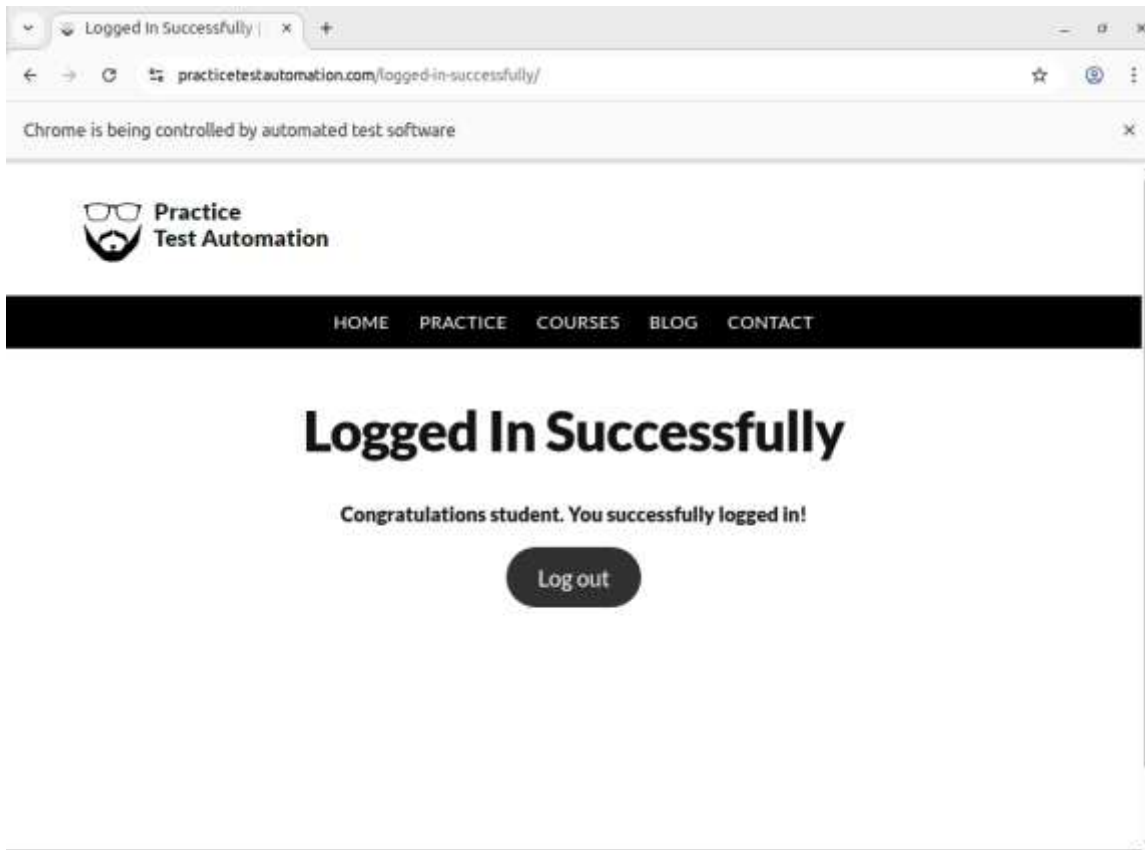
[INFO] -----  

[INFO]
[INFO] --- maven-resources-plugin:3.3.1:resources (default-resources) @ MyMavenSeleniumApp03-1.0-SNAPSHOT ---
[INFO] Copying 0 resource to /target/classes from /src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.10.1:compile (default-compile) @ MyMavenSeleniumApp03-1.0-SNAPSHOT ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-surefire-plugin:3.0.0-M5:surefire-test (default-test) @ MyMavenSeleniumApp03-1.0-SNAPSHOT ---
[INFO]
[INFO] --- maven-jar-plugin:3.3.0:jar (default-jar) @ MyMavenSeleniumApp03-1.0-SNAPSHOT ---
[INFO] Building jar: /target/MyMavenSeleniumApp03-1.0-SNAPSHOT.jar
[INFO]
[INFO] -----  

[INFO] BUILD SUCCESS
[INFO] -----
[INFO]

```





Sauce Demo Website

Sauce Demo (<https://www.saucedemo.com/>) is a fake e-commerce website created by Sauce Labs for testing and learning web automation tools like Selenium.

It is used for:

- Writing automation scripts and UI tests
- Demonstrating cloud testing features
- Learning and practicing test automation

Practice Test Automation Website

<https://practicetestautomation.com/practice-test-login/> is a sample website used for testing and learning web automation tools like Selenium.

It is used for:

- Writing automation scripts and UI tests
- Practicing login functionality testing
- Learning and improving test automation skills

Automation Exercise Website

<https://automationexercise.com/products> is a sample website used for testing and learning web automation tools like Selenium.

It is used for:

- Writing automation scripts and UI tests
- Practicing login functionality testing
- Learning and improving test automation skills

Pushing The Project into Git:

Execute the commands:

```
$ git config --global user.name "vaishnavim1121"
```

```
$ git config --global user.email "vaishnavim1121@gmail.com"
```

```
$git init
```

```
$git add .
```

```
$git commit -m "initial commit"
```


\$ git remote add origin <https://github.com/vaishnavim1121/MyMavenSeleniumApp.git>

\$ git remote set-url origin https://*PAT*@github.com/vaishnavim1121/MyMavenSeleniumApp

\$ git push -u origin master

```

Total time: 22.748 s
Finished at: 2026-03-04T22:25:25+05:30
-----
vaishnav@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp$ git config --global user.name "vaishnavim1121"
vaishnav@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp$ git config --global user.email "vaishnavim1121@gmail.com"
vaishnav@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'Trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/vaishnavi/devops21/MyMavenSeleniumApp82/.git/
vaishnav@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp$ git add .
vaishnav@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp$ git commit -m "Initial"
[master (root-commit) 72300fc] Initial
13 files changed, 204 insertions(+)
create mode 100644 pom.xml
create mode 100644 src/main/java/com/example/App.java
create mode 100644 src/test/java/com/example/AppTest.java
create mode 100644 target/MyMavenSeleniumApp2-1.0-SNAPSHOT.jar
create mode 100644 target/classes/com/example/App.class
create mode 100644 target/maven-archiver/pom.properties
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
create mode 100644 target/surefire-reports/TEST-com.example.AppTest.xml
create mode 100644 target/surefire-reports/com.example.AppTest.txt
create mode 100644 target/test-classes/com/example/AppTest.class
vaishnav@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp$ git remote add origin https://github.com/vaishnavim1121/MyMavenSeleniumApp82.git
vaishnav@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp$ git remote set-url origin https://ghp_9WvYmP05agt2QxFOV7nGjt3t4L6ZV3wVIFI@github.com/vaishnavim1121/MyMavenSeleniumApp82
vaishnav@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp$ git push -u origin master
Enumerating objects: 39, done.
Counting objects: 100% (39/39), done.
Delta compression using up to 16 threads
Compressing objects: 100% (19/19), done.
Writing objects: 100% (39/39), 8.51 KiB | 1.21 MiB/s, done.

```